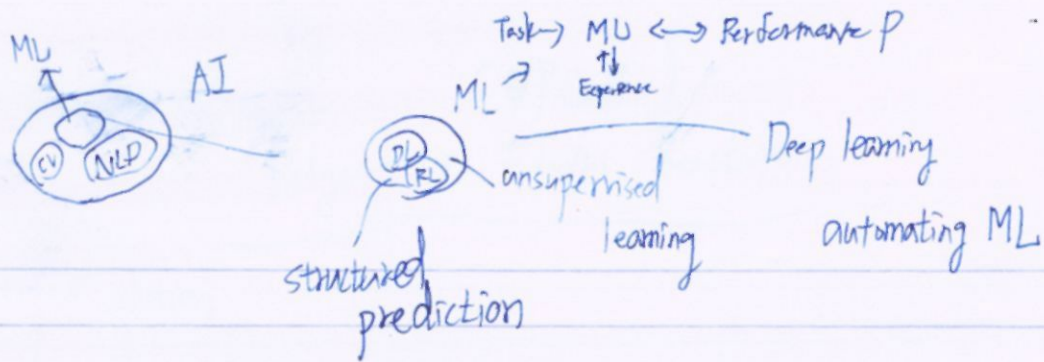




Facial recognition

Recommendation system

before

rule-based translation (~2000)

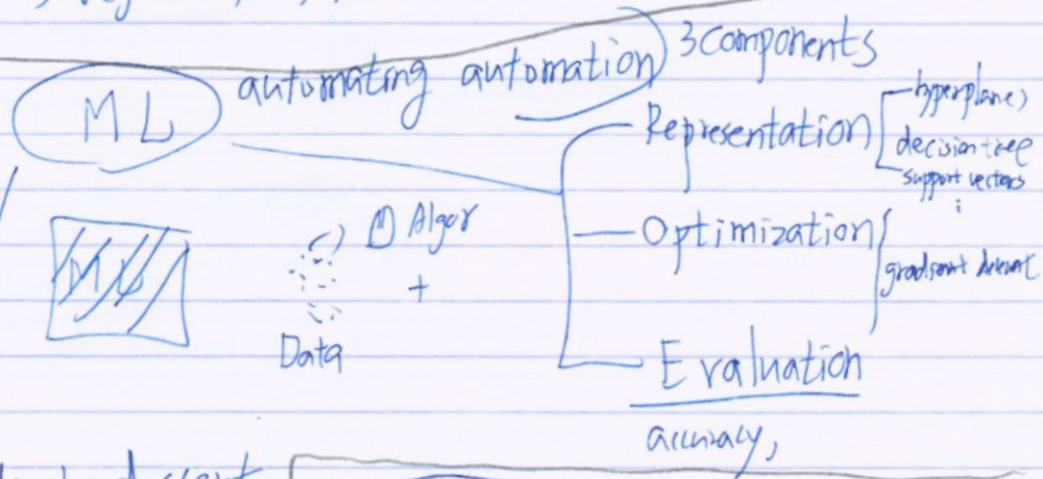
$\rightarrow$ ML

Medical diagnosis

k-nearest Neighbors $\begin{cases} \text{classification} \\ \text{regression} \end{cases}$

ML basics + linear algebra review
 HW: income classification

linear classification
 regression $\rightarrow$ DL



gradient descent

learning rate

partially observed learning:

learning setting

supervised learning $f(x)=y$
 class: input: picture, out: type
 regression: input: test, out: value

unsupervised learning: 分类, 降维 only
 keep important structure

semi-supervised learning

Output y : regression
e.g. price
or
classification

$D = \{(x_i, y_i)\}_{i=1}^n$ scenario:
① changing frequently

Supervised Learning

$$f(x_i) = y_i$$

or
structured

e.g. Semantic segmentation

classification
(binary, multiclass, structured)

1. set up a supervised learning problem 3. Representation $x \rightarrow$ proper form
2. Data collection (x, y) 4. Modeling: hypothesis space
5. Learning 6. Model Selection 7. go or repeat

Goal: min loss function

by e.g. gradient descent

GD: $\begin{cases} \text{Batch} \\ \text{Stochastic} \\ \text{mini-batch} \end{cases}$

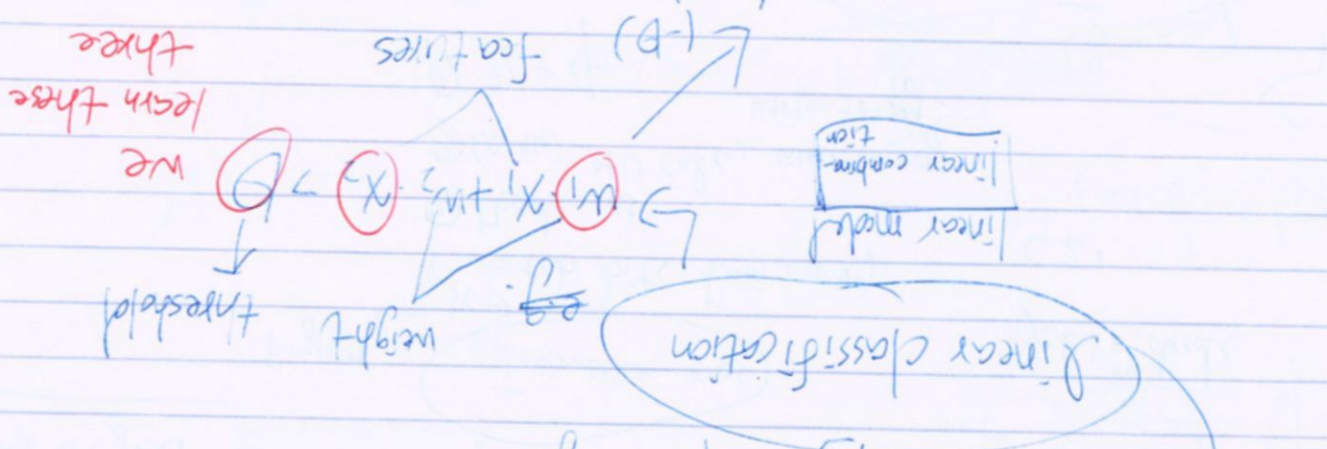
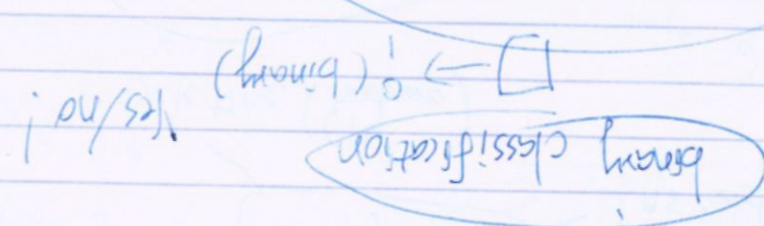
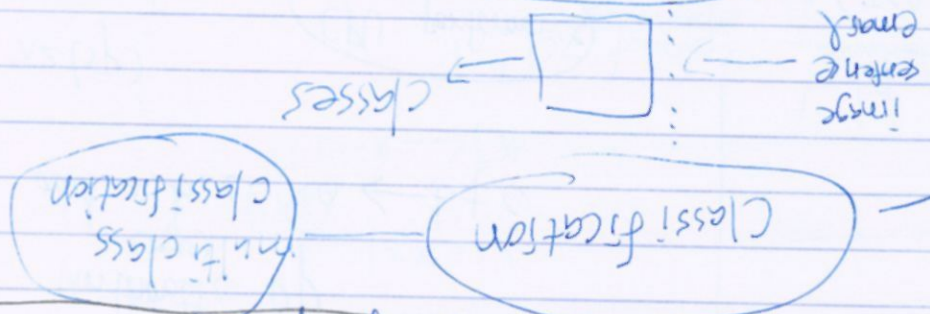
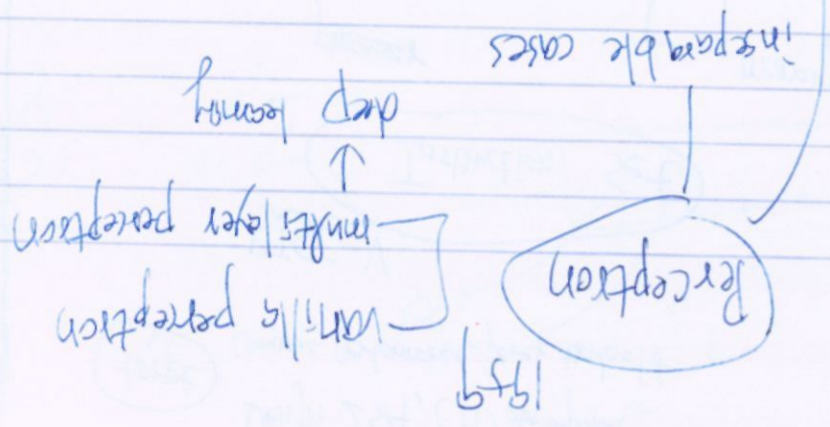
If loss function is convex, gd is guaranteed to converge to the global min

target: $y = w^T x + \epsilon$ ^{noise}

Linear regression

$\begin{cases} \text{MSE} \\ \text{MAE} \\ \text{Huber loss} \end{cases}$

How to define different M ?



$$w_1 \cdot x_1 + w_2 \cdot x_2 + b > 0$$

bias term

$$\Rightarrow w_0 \cdot 1 + w_1 \cdot x_1 + w_2 \cdot x_2 > 0$$

$d+1$

$1 \cdot x_0$

$\Rightarrow \frac{w}{x} > 0$

$\Rightarrow \frac{w}{x} > 0$

$f(x) = a(w \cdot x)$

$s > 0$

-1 other

Perceptron Algor

1. $\vec{w} \leftarrow (0, 0, 0 \dots 0) \Rightarrow \text{output: } \checkmark$

2. find error $y \cdot (\vec{w} \cdot \vec{x}) \leq 0$

3. update $\vec{w} \Rightarrow \vec{w}' \leftarrow \vec{w} + \vec{x}$ (for positive case)

$$\vec{w}' \cdot \vec{x} = \vec{w} \cdot \vec{x} + \underbrace{\vec{x} \cdot \vec{x}}_{> 0} > \vec{w} \cdot \vec{x} \quad \vec{w}' \leftarrow \vec{w} - \vec{x} \text{ (neg)} \\ \Rightarrow \vec{w}' \leftarrow \vec{w} + y \cdot \vec{x}$$

4. Convergence $\rightarrow$ (best case) in a round, all examples are classified correctly by the $\vec{w}$

$\Rightarrow$ (General) $\circ$ max limit iteration

$\oplus$ dev set decide the best iteration

Linear separation $\Rightarrow$ Perceptron convergence theorem

support $\rightarrow$ SVM find the line that $\max \gamma$ (linear buffer zone)

D has γ (linear separable)

$\Rightarrow$ perceptron algorithm must CONVERGE before $\frac{R^2}{\gamma^2}$

$$\text{mistakes (Updates)} \leq \frac{R^2}{\gamma^2} \quad (R = \max_{(\vec{x}, y) \in D} \|\vec{x}\|)$$

Handle Inseparable case

original feature space $\xrightarrow{\text{project}}$ higher-dimensional feature space

$\Rightarrow$ (possible) linear separable

$\leftarrow$ Kernel trick of SVM ... methods

epoch: 完整训练一次
 $\Phi = \phi = f \circ \phi$
 大, 小
 大, 小
 大, 小

$$\text{Prediction} = \text{sign} \left(\sum_{k=1}^n c_k \text{sign}(w_k \cdot x_{\text{test}}) \right)$$

improvement of vanilla perceptron
 voted perceptron
 averaged perceptron
 ① much less sensitive
 ② large margin

mistakes (updates) $\leq \frac{R^2(D)}{\delta^2(D)}$
 $\|w\| = 1$
 oracle writer

$\Phi(\cdot) = \text{feature map}$
 $y(n \cdot \Phi(x)) \geq \delta, \text{ for all } (x, y) \in D$
 $\Rightarrow$ linearly separable + feature map

In general,
 $\hookrightarrow$ separated by the plane $y=0 \rightarrow$ got nonlinear decision boundary
 in low dimensions
 project back

odd $x_1, x_2, x_1 x_2$
 feature transformation
 kernel method

$$y = x_1 x_2 \Rightarrow$$

x_1	x_2	y
+	+	+
+	-	-
-	+	-
-	-	+

XOR example
 not linearly separable

Regression

target: $\min[\text{diff}(\text{prediction}, \text{truth})]$

measured by $\rightarrow$ mean squared error (MSE)

$$y = w \cdot s + b \Rightarrow y = \underbrace{w_1 \cdot s + w_2 \cdot r + b}_{\text{match target}}$$

$\downarrow$ finally
make predictions on new data

(features)
 $X = (x_0=1, x_1, x_2, \dots, x_d) \Rightarrow d+1$ dimensional augmented space

$$\vec{w} = (w_0, w_1, w_2, \dots, w_d)$$

$$\Rightarrow \hat{y} = \vec{w} \cdot \vec{x}, \hat{y} = \text{predicted value}$$

是2還是n沒影響

量化成果 $\Rightarrow$ cost function: $J(\vec{w}) = \frac{1}{2} \sum_{(\vec{x}, y) \in D} (f_{\vec{w}}(\vec{x}) - y)^2$
as small as possible

Regularization

$\lambda > 0$ (hyperparameter)

optimize (min)
iteratively

$$L_2: J(\vec{w}) = \frac{1}{2} \sum_{(\vec{x}, y) \in D} (f_{\vec{w}}(\vec{x}) - y)^2 + \lambda \|\vec{w}\|_2^2$$

aka.
regularization
strength

$$\lambda (\sqrt{w_0^2 + w_1^2 + w_2^2 + \dots + w_n^2})^2$$

optimize (min)

$$L_1: J(\vec{w}) = \frac{1}{2} \sum_{(\vec{x}, y) \in D} (f_{\vec{w}}(\vec{x}) - y)^2 + \lambda \|\vec{w}\|_1$$

$$\lambda (|w_0| + |w_1| + |w_2| + \dots)$$

Gradient Descent

optimize (min) cost function iteratively

or

LMS algr. $\vec{w} \leftarrow \vec{w} + \alpha (y - \vec{w} \cdot \vec{x}) \vec{x}$ (one example)
(least mean squares)

more data's

$$\Rightarrow \vec{w} \leftarrow \vec{w} + \alpha \sum_{(\vec{x}, y) \in D} (y - \vec{w} \cdot \vec{x}) \vec{x}$$

每次更新都是需要完整的数据 set

Normal Equation

for large-scale datasets

SGD

Stochastic Gradient Descent

BGD

simplify

$$\vec{w}_i \leftarrow \vec{w} + \alpha \sum_D (y - \vec{w} \cdot \vec{x}) \vec{x} \implies \vec{w}' \leftarrow \vec{w} + \alpha (y^{(i)} - \vec{w} \cdot \vec{x}^{(i)}) \vec{x}^{(i)}$$

randomly picked from D

faster convergence but less precise

stability $\longleftrightarrow$ efficiency

$\Downarrow$

Mini-Batch Gradient Descent

Normal Equation

another way to $\min J(\vec{w})$